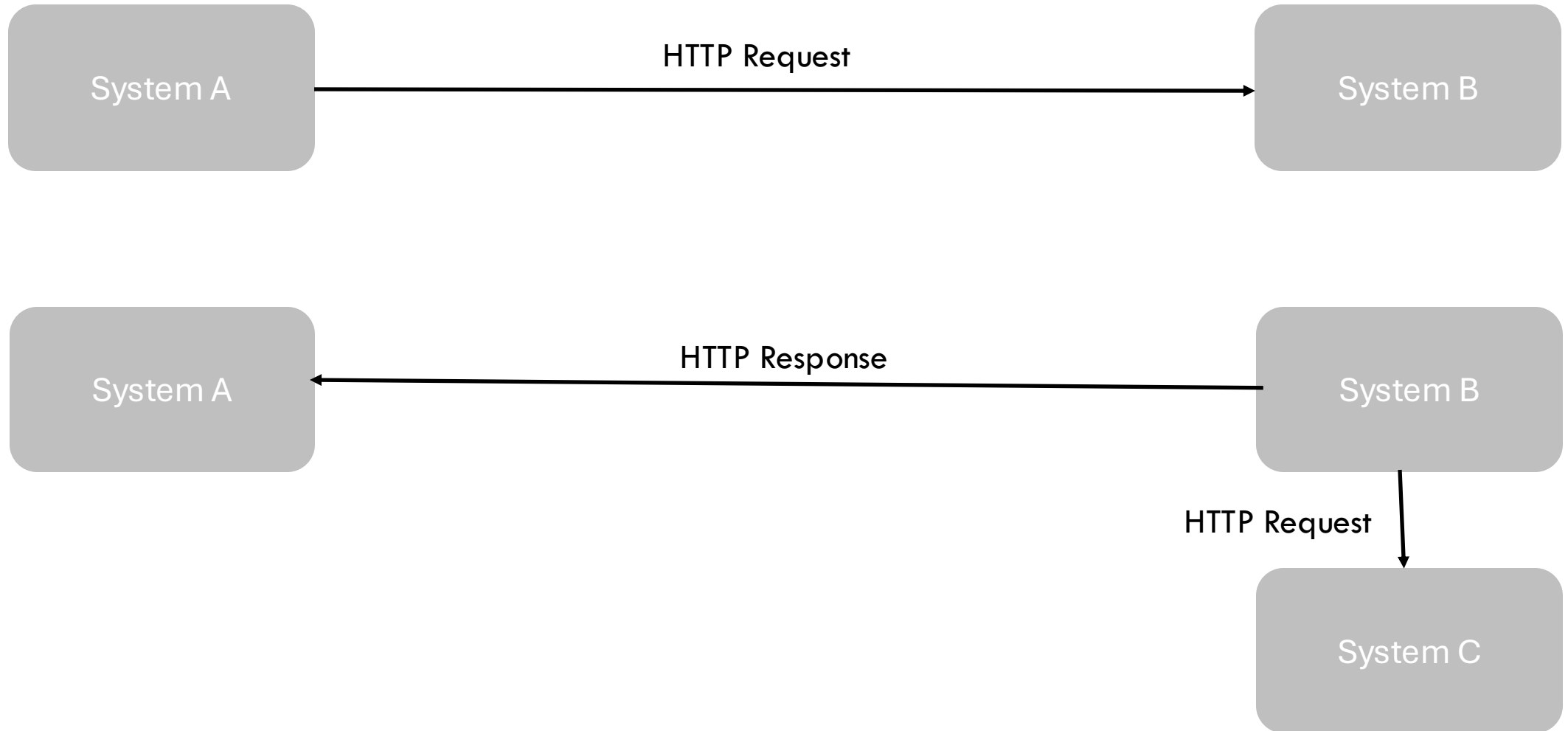
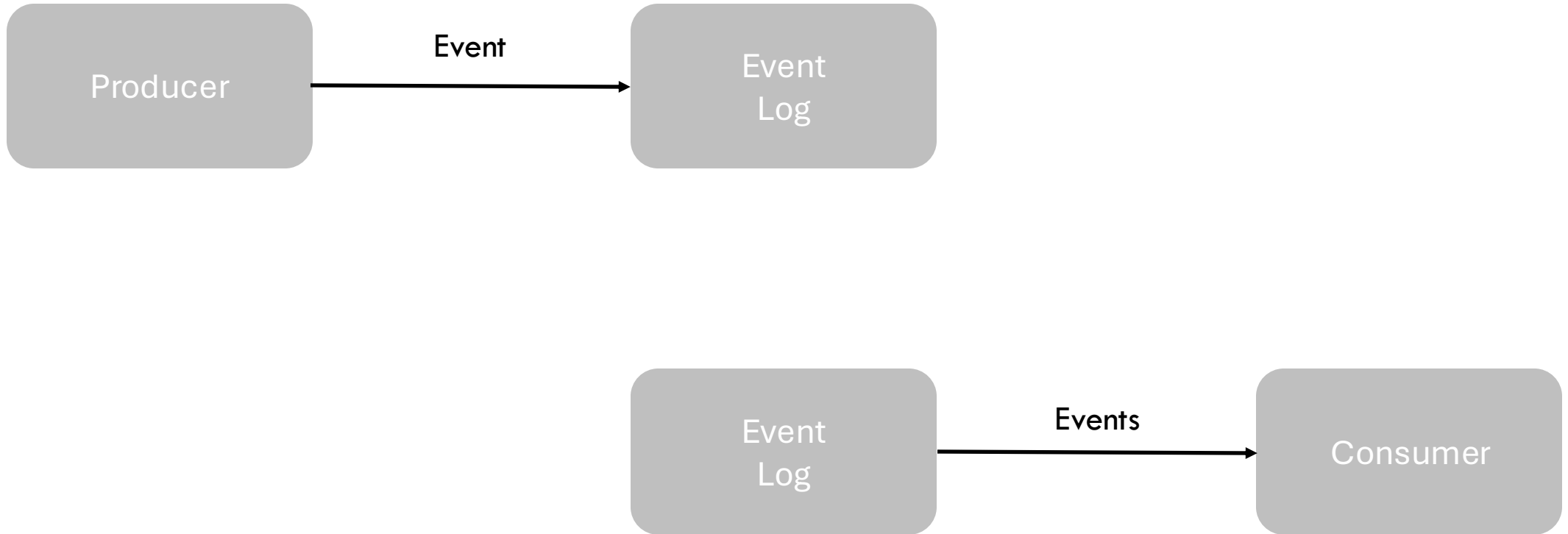


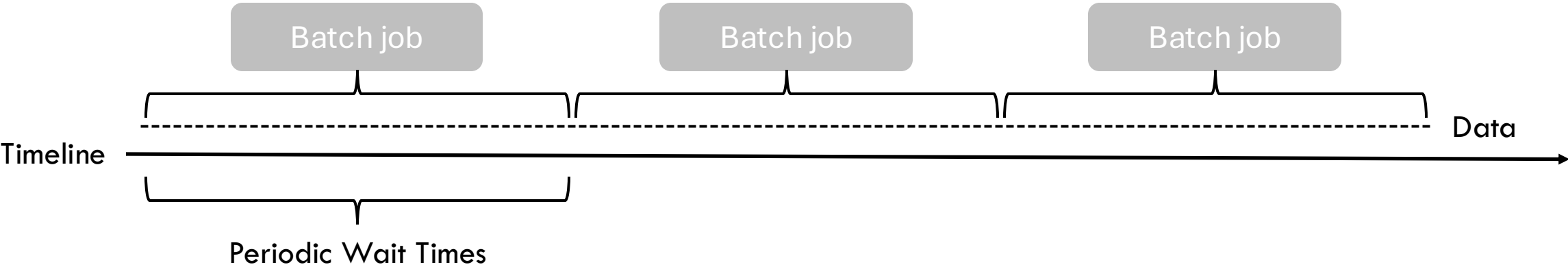
Traditional API



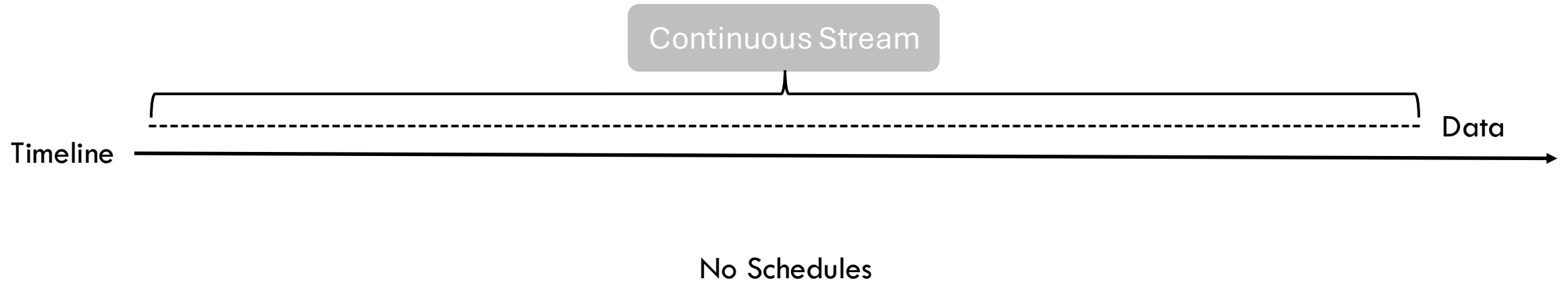
Append-Only Log



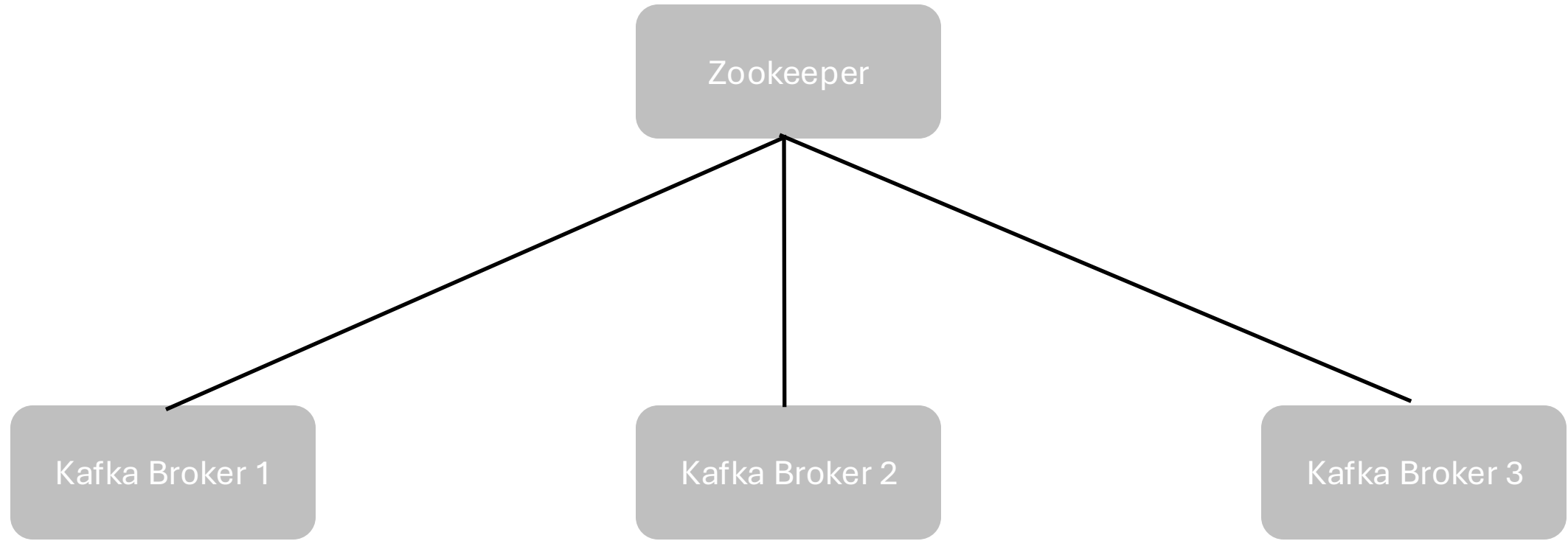
Batch-Processing Timeline



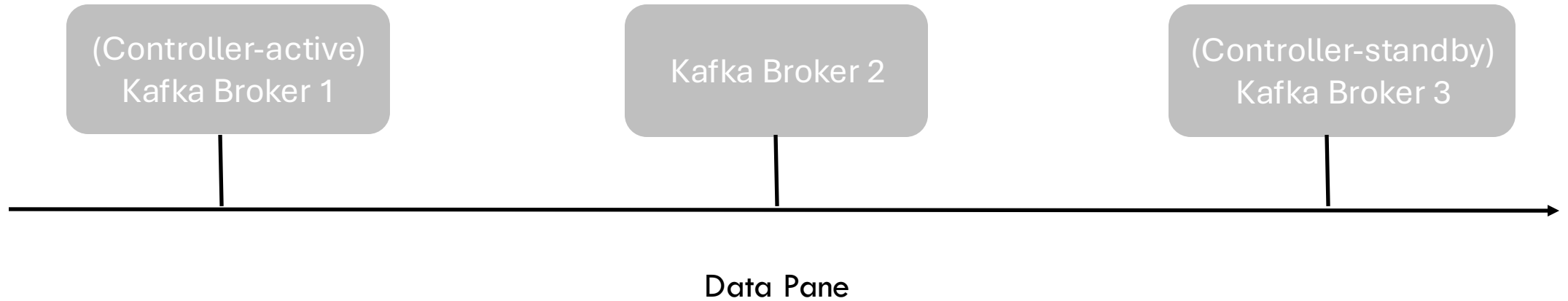
Stream-Processing Timeline



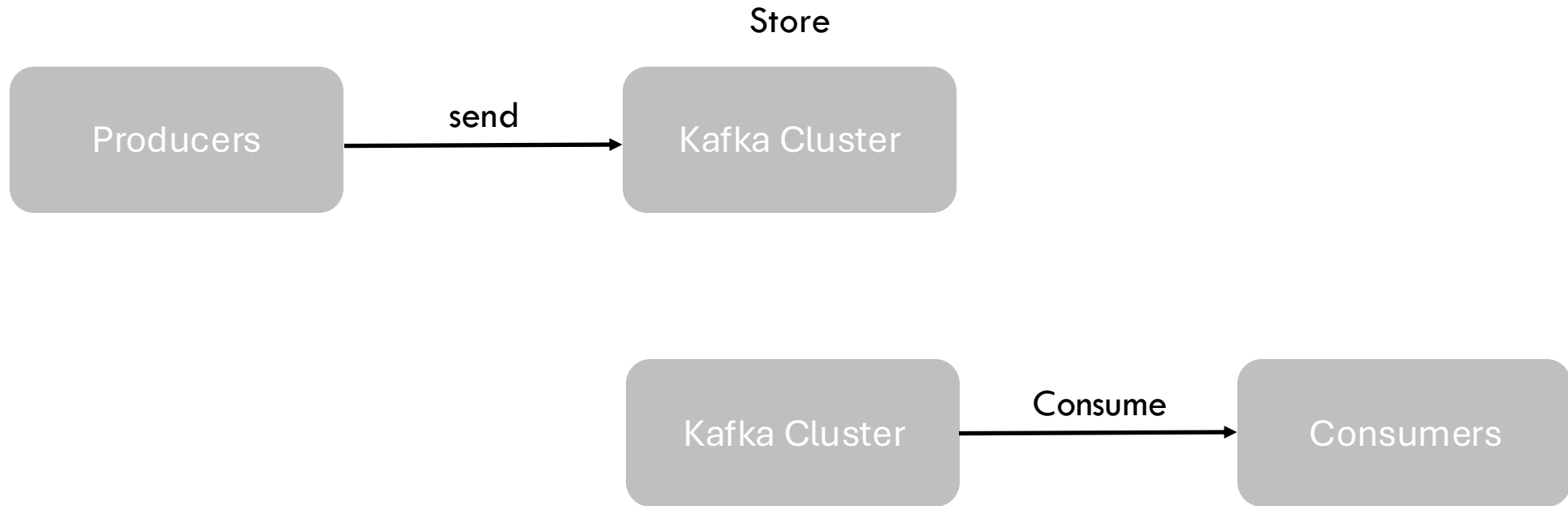
Legacy Architecture (Kafka + ZooKeeper)



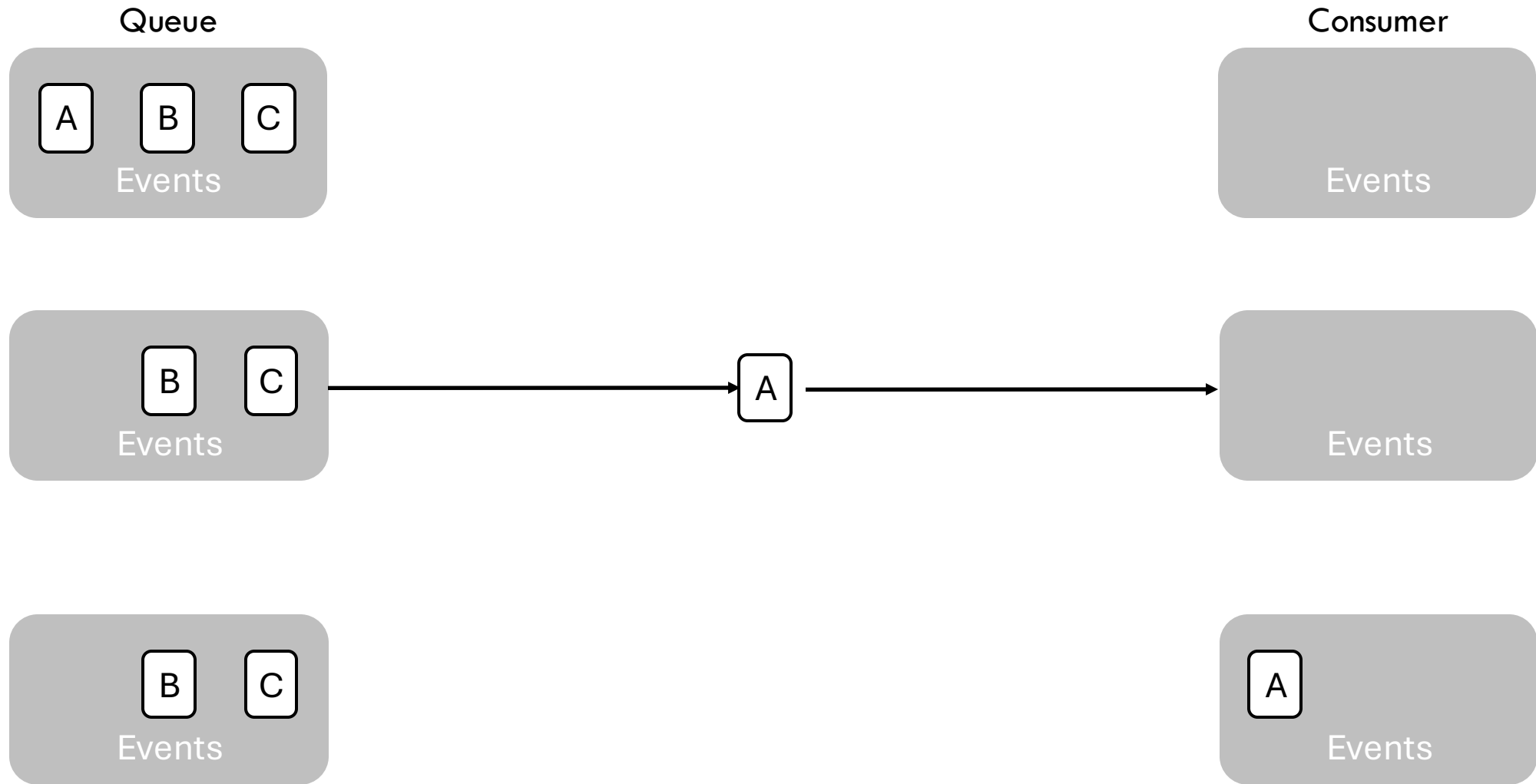
KRaft Architecture



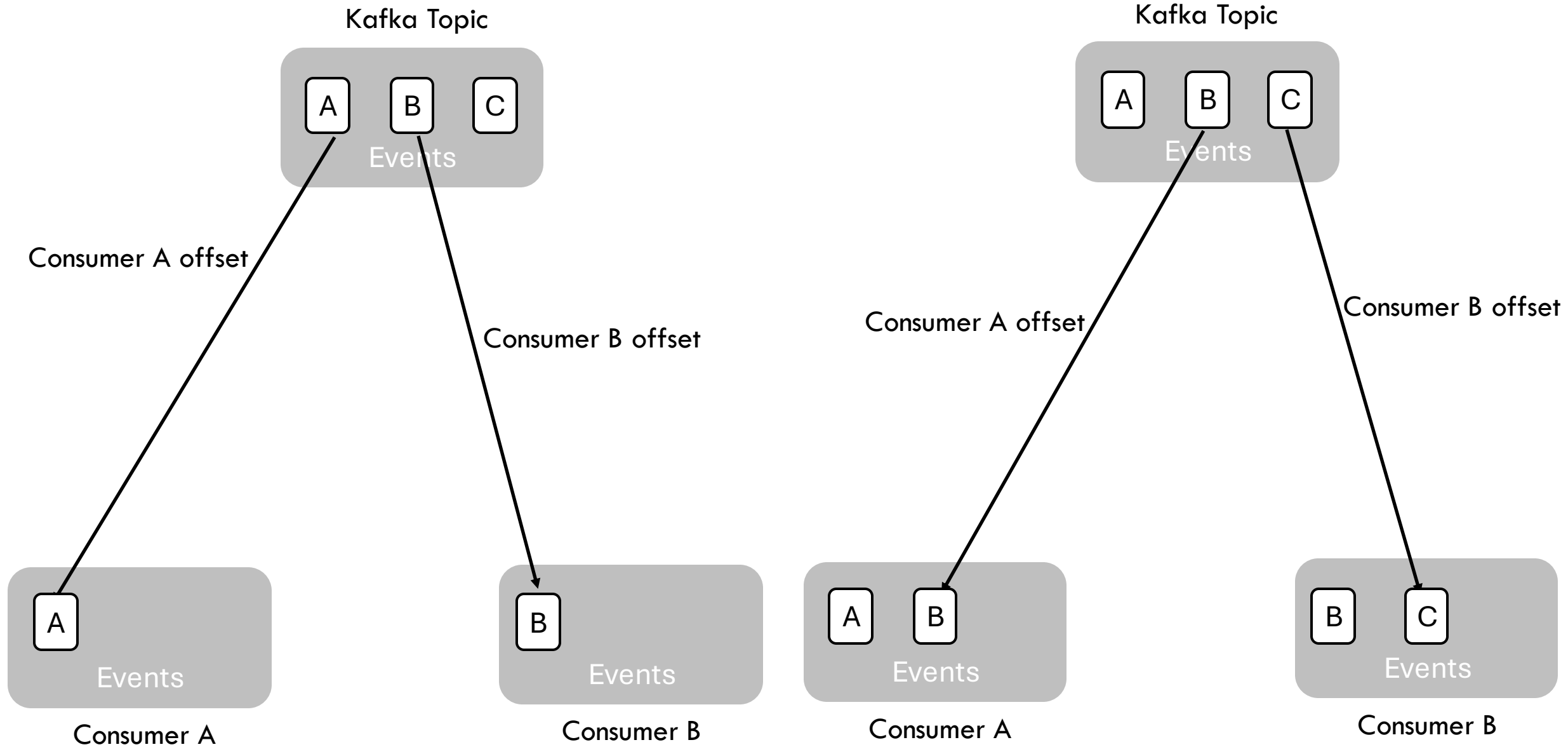
Kafka



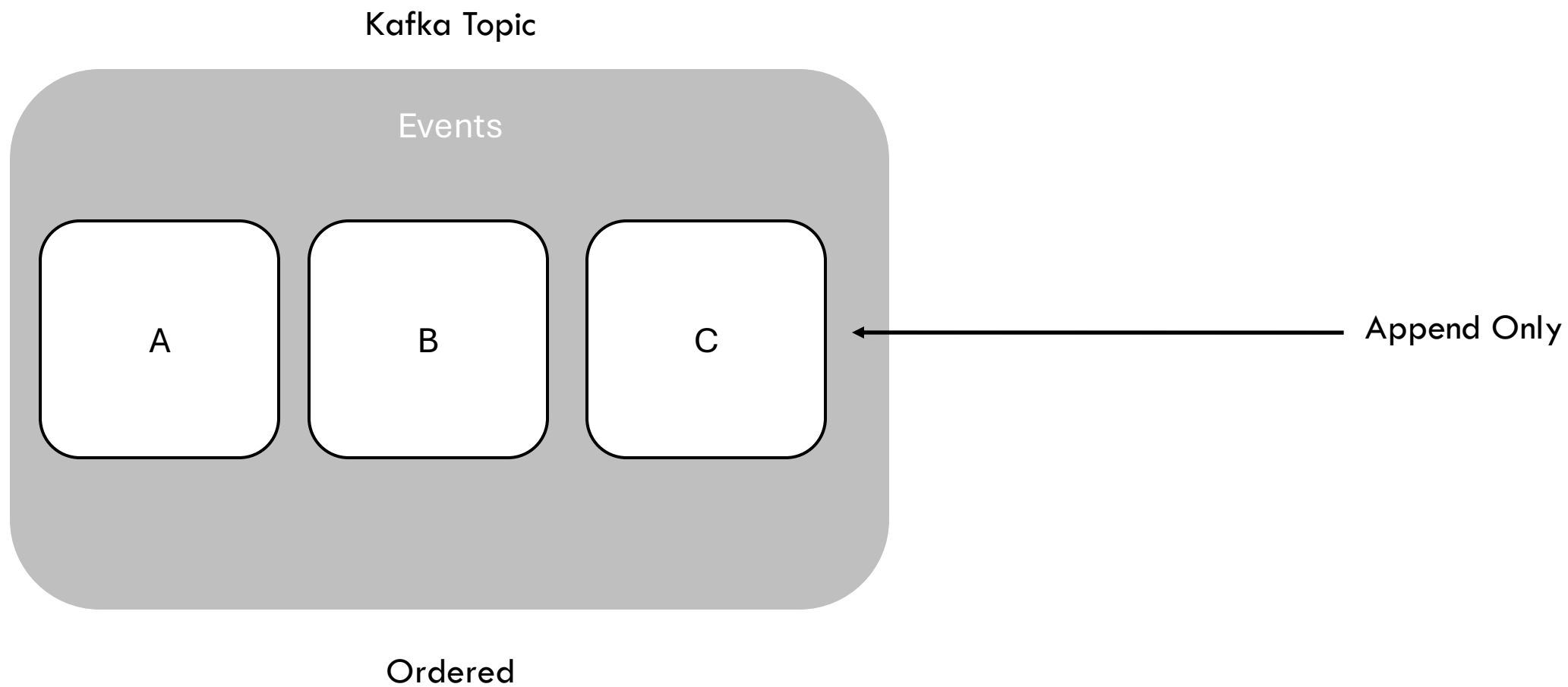
Traditional Queue



Kafka-style log



Kafka Topic



Partitions

Kafka Topic

Partitions
Partition 2

Partition 1

Partition 3

Event A

Event C

Event F

Events Flow



Event B

Event E

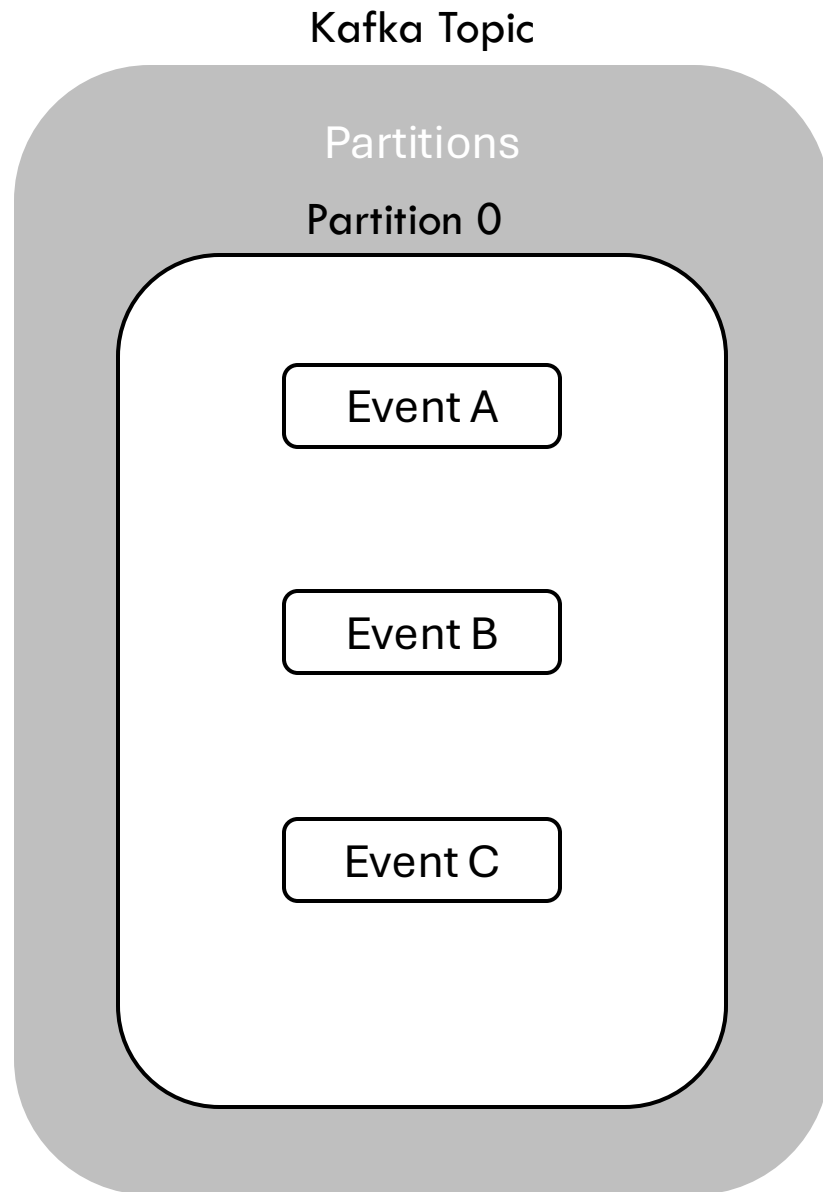
Event G

Event D

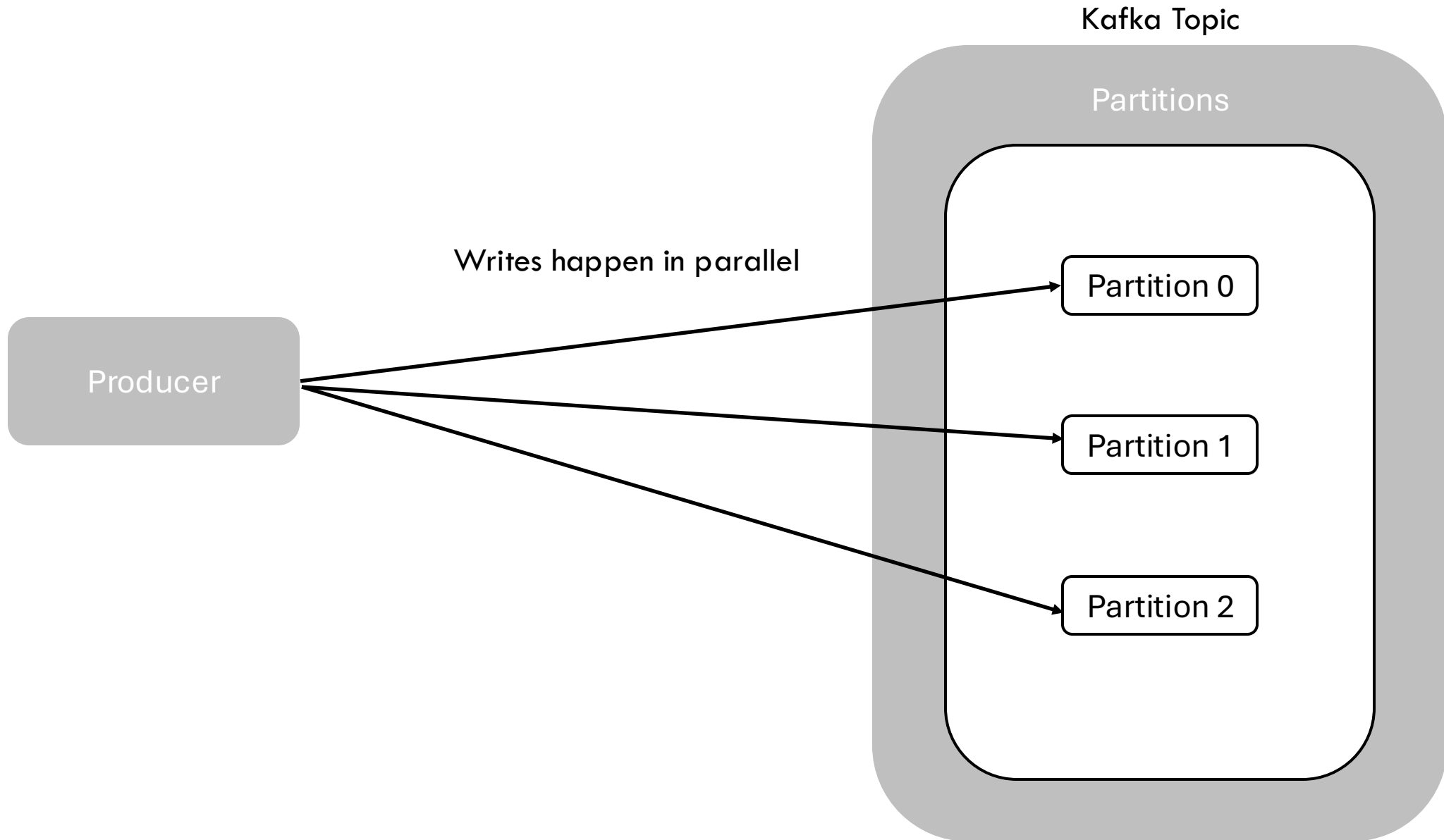
Event H

Event I

Partitions



Producer



Consumer

Kafka Topic

Partitions

Partition 0

Offset:1

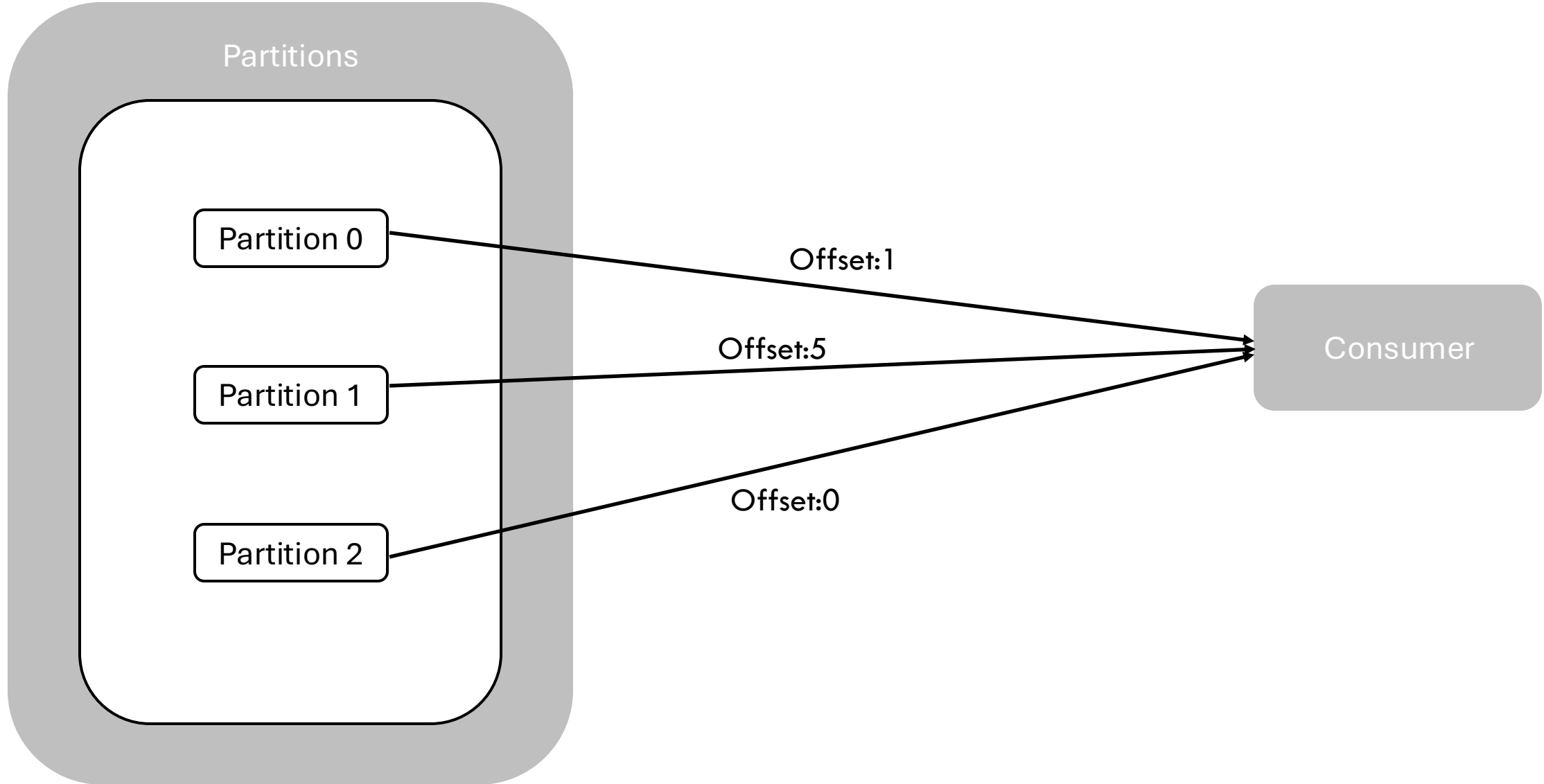
Partition 1

Offset:5

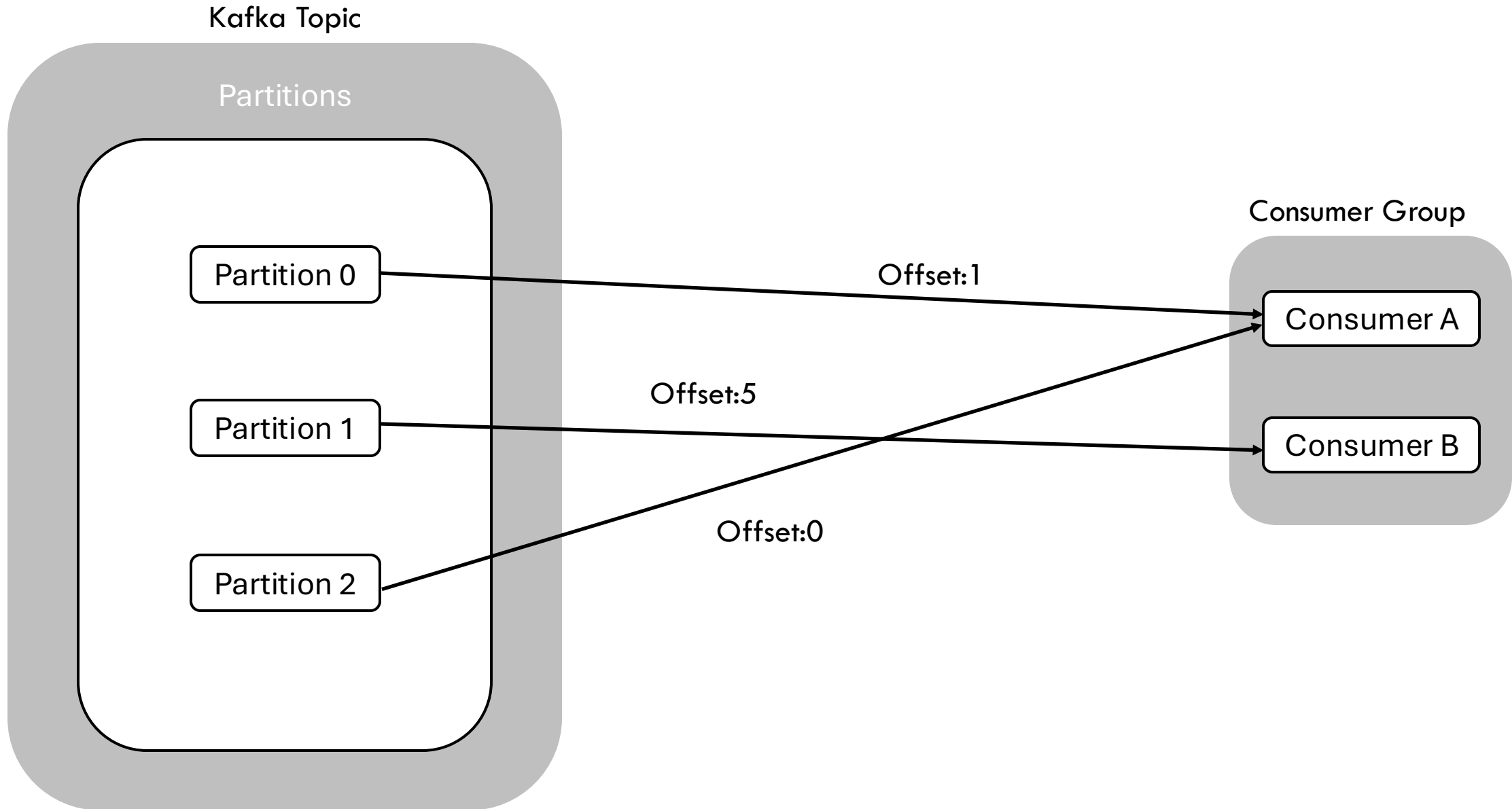
Partition 2

Offset:0

Consumer



Consumer Group



Components

Controller(active)

Broker-1

Topic(orders)

Partition 0 (leader)

Partition 1 (follower)

Topic(users)

Partition 0(follower)

Partition 1(leader)

Broker-2

Topic(orders)

Partition 0 (follower)

Partition 1 (follower)

Topic(users)

Partition 0 (leader)

Partition 1(follower)

Controller(standby)

Broker-3

Topic(orders)

Partition 0 (follower)

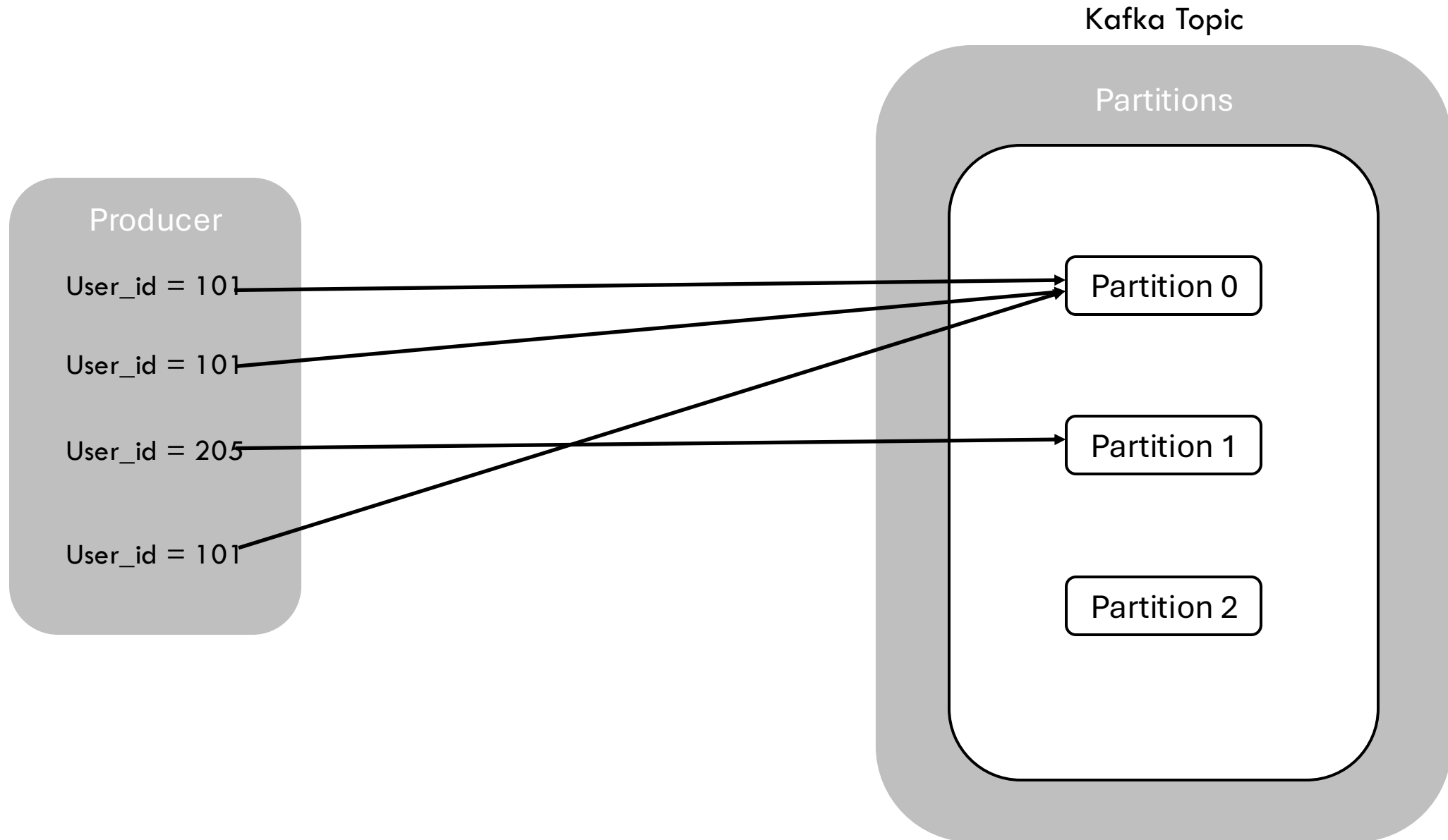
Partition 1 (leader)

Topic(users)

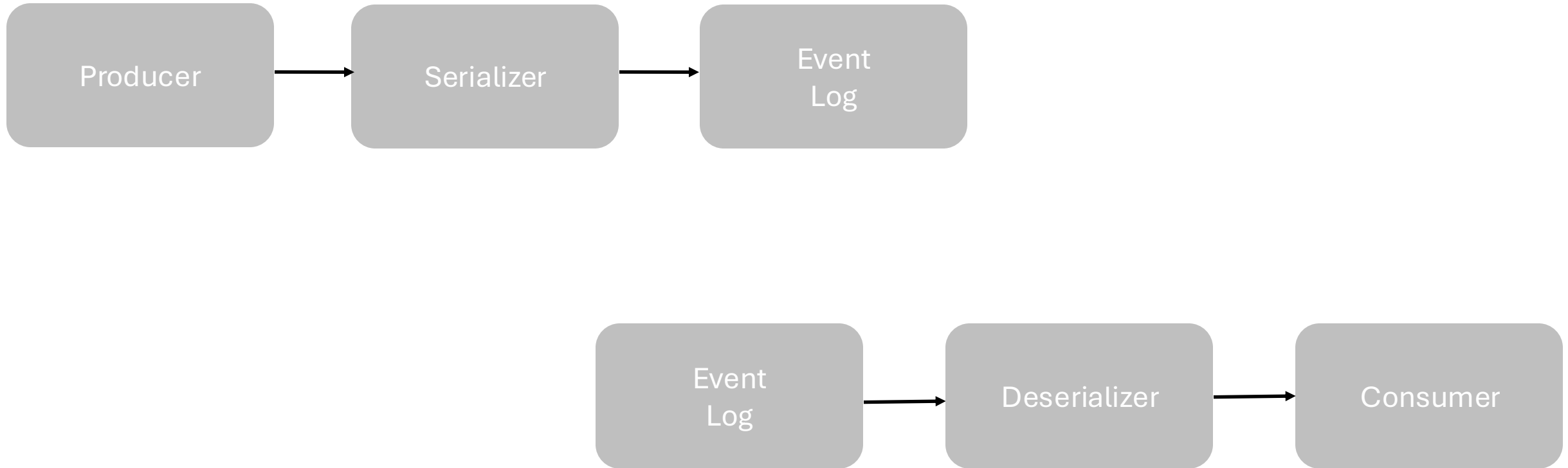
Partition 0(follower)

Partition 1(follower)

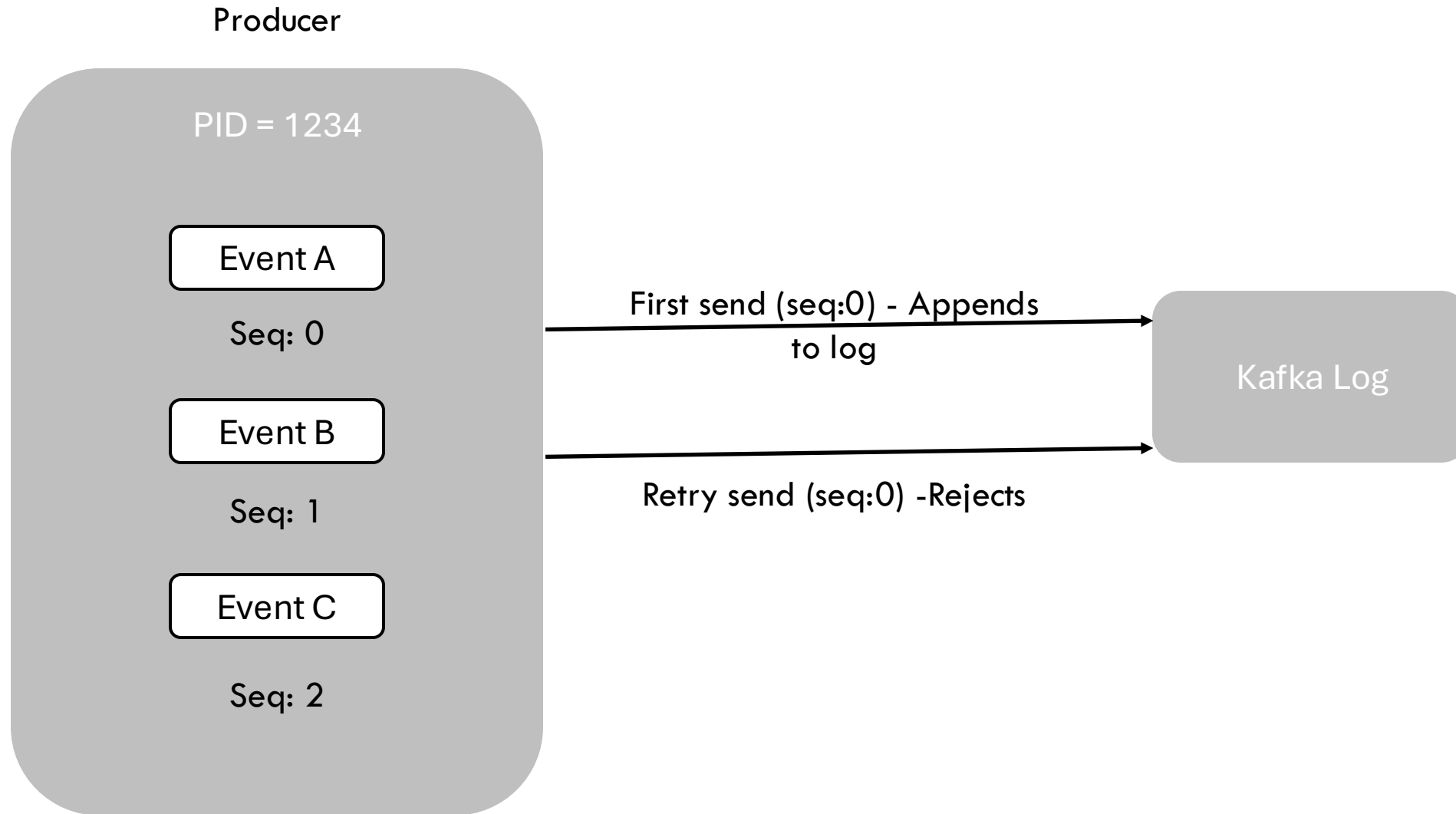
Message Keys



Serialization

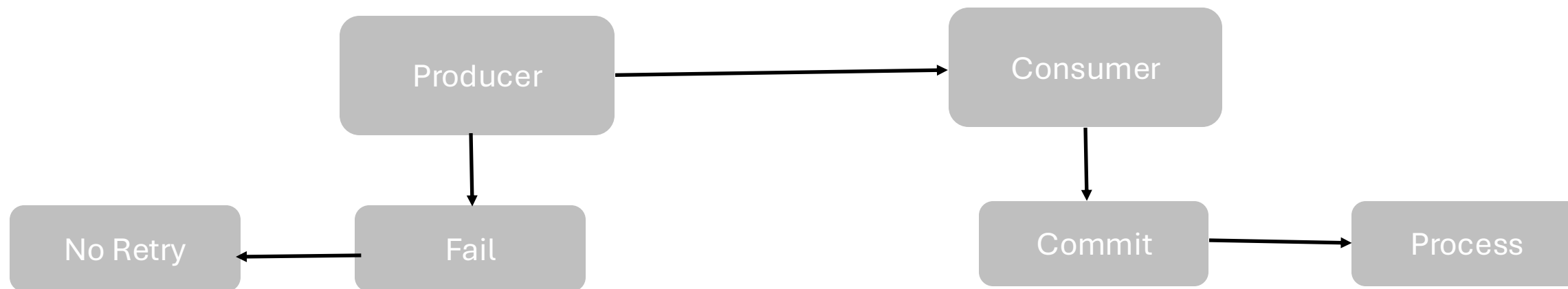


Idempotency



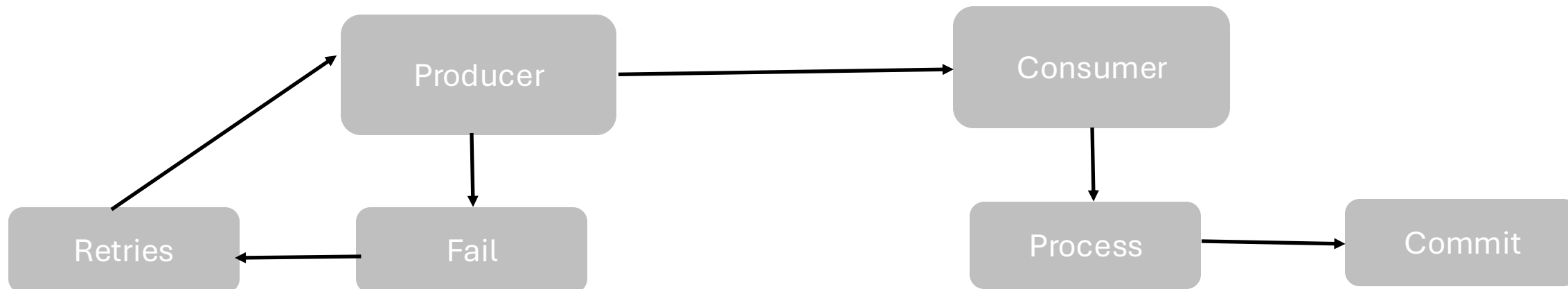
Delivery Semantics

At-most-once



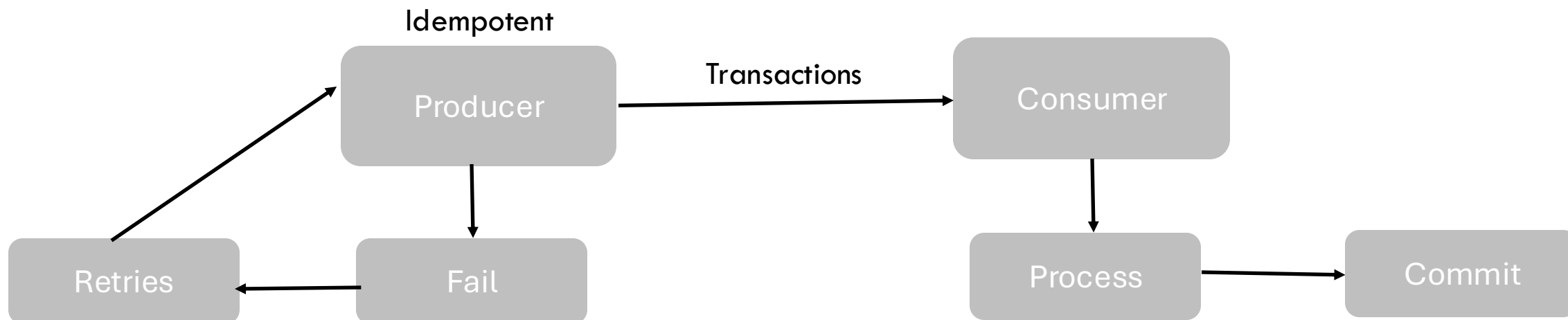
Delivery Semantics

At-least-once

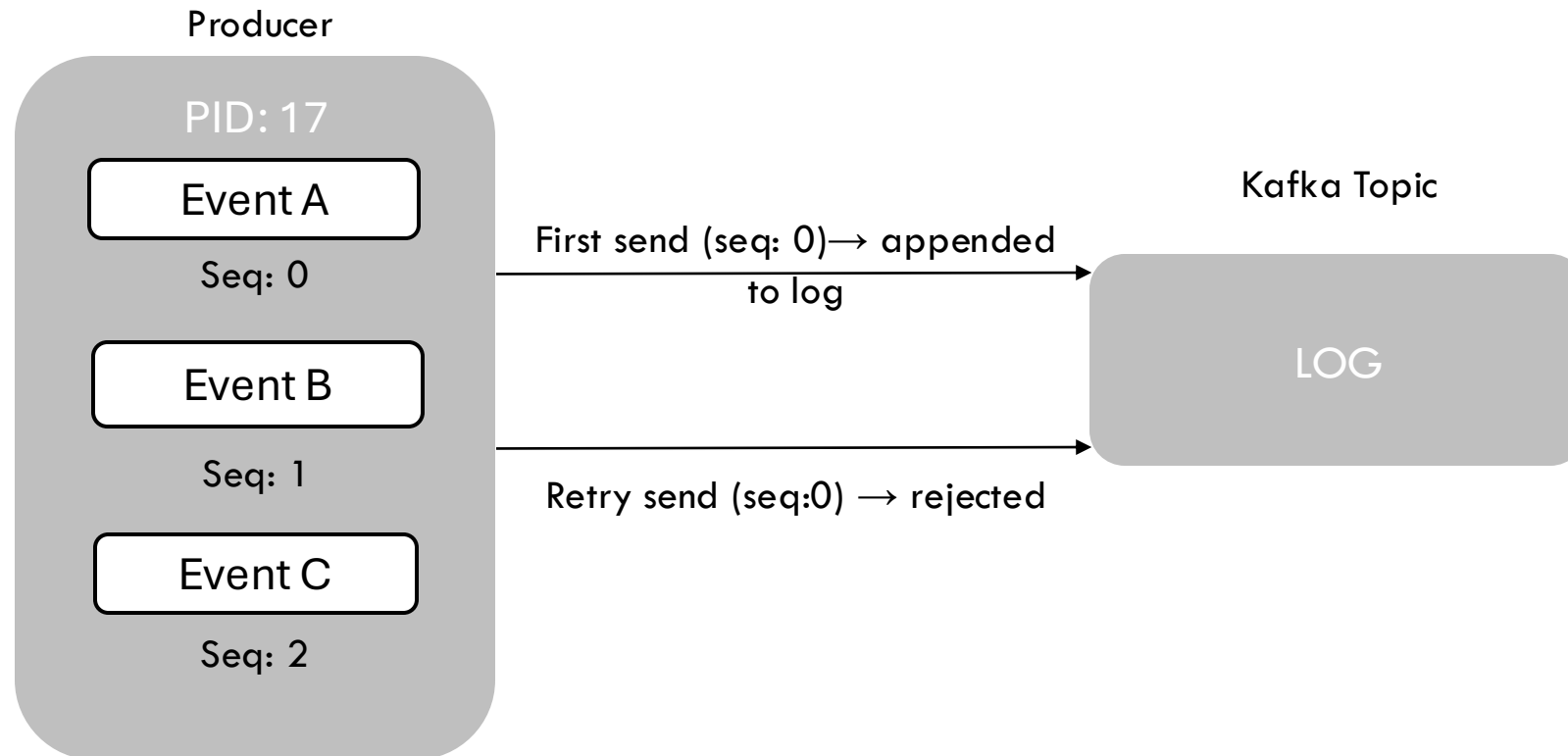


Delivery Semantics

Exactly-once

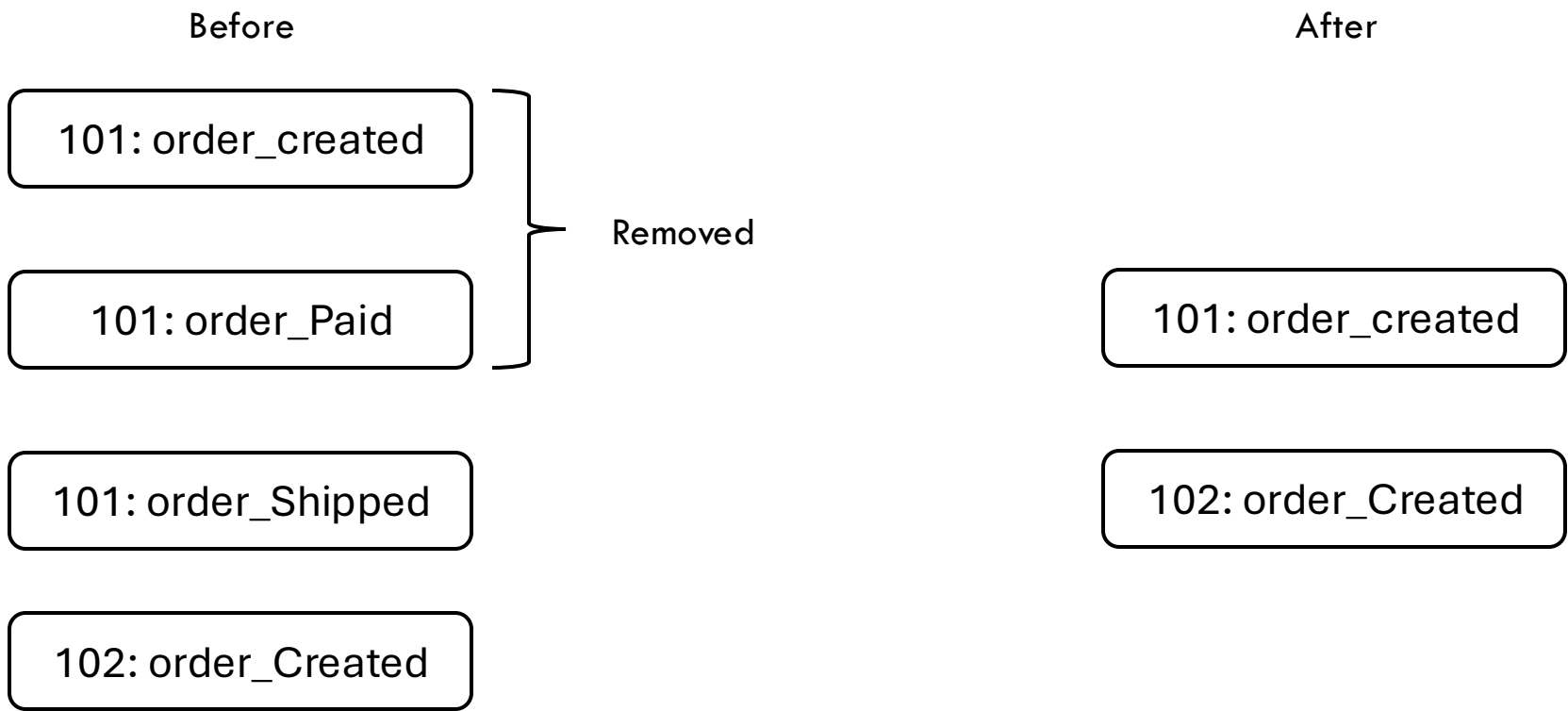


Idempotency

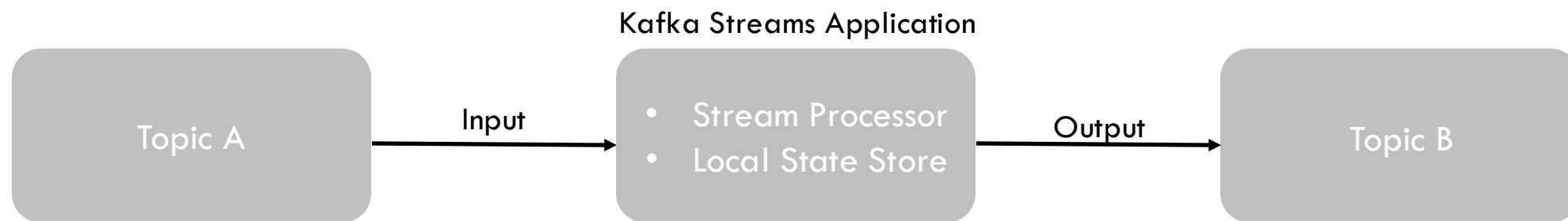


Compact Log

Cleanup Policy = Compact



Kafka Streams

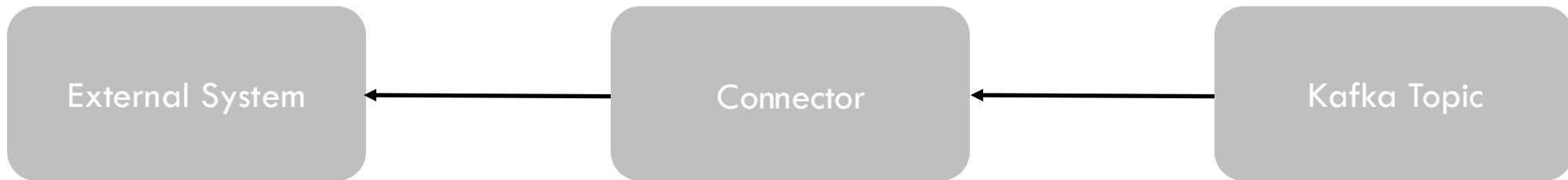


Kafka Connect

Source Connector



Sink Connector



Kafka Scaling

Kafka Cluster

Consumer Group

Broker 1

Broker 2

Broker 3

P0

P1

P2

P3

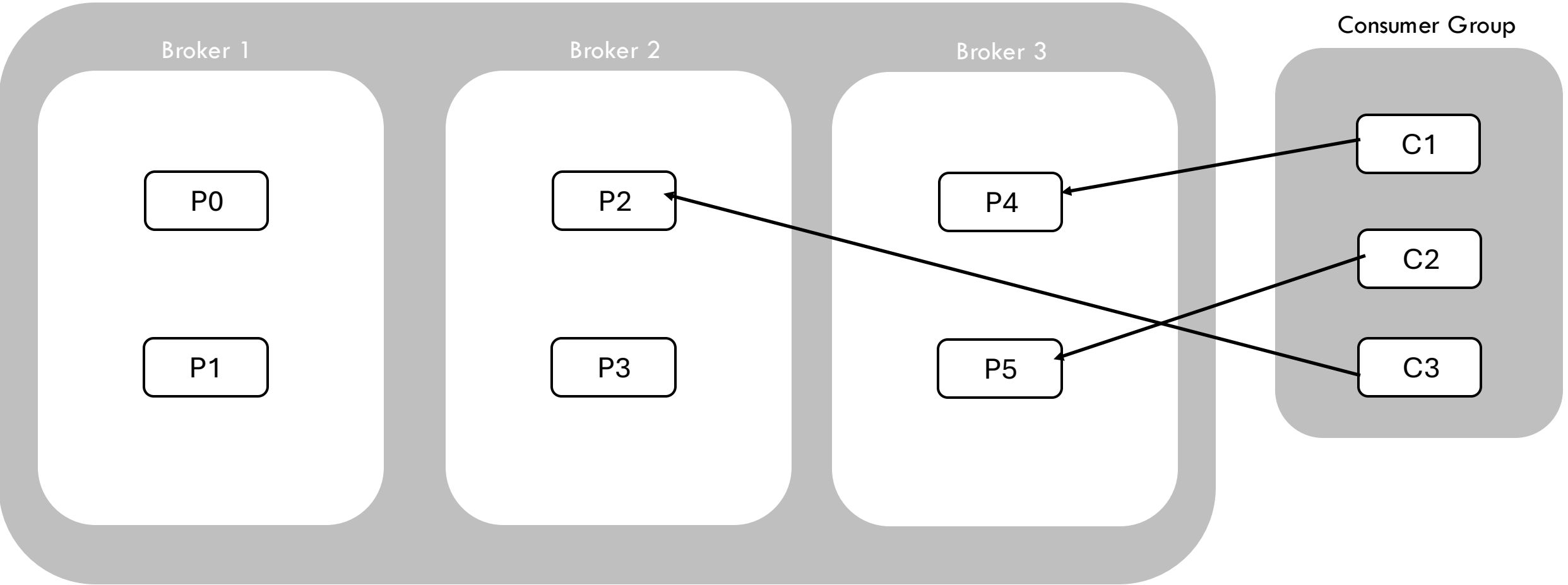
P4

P5

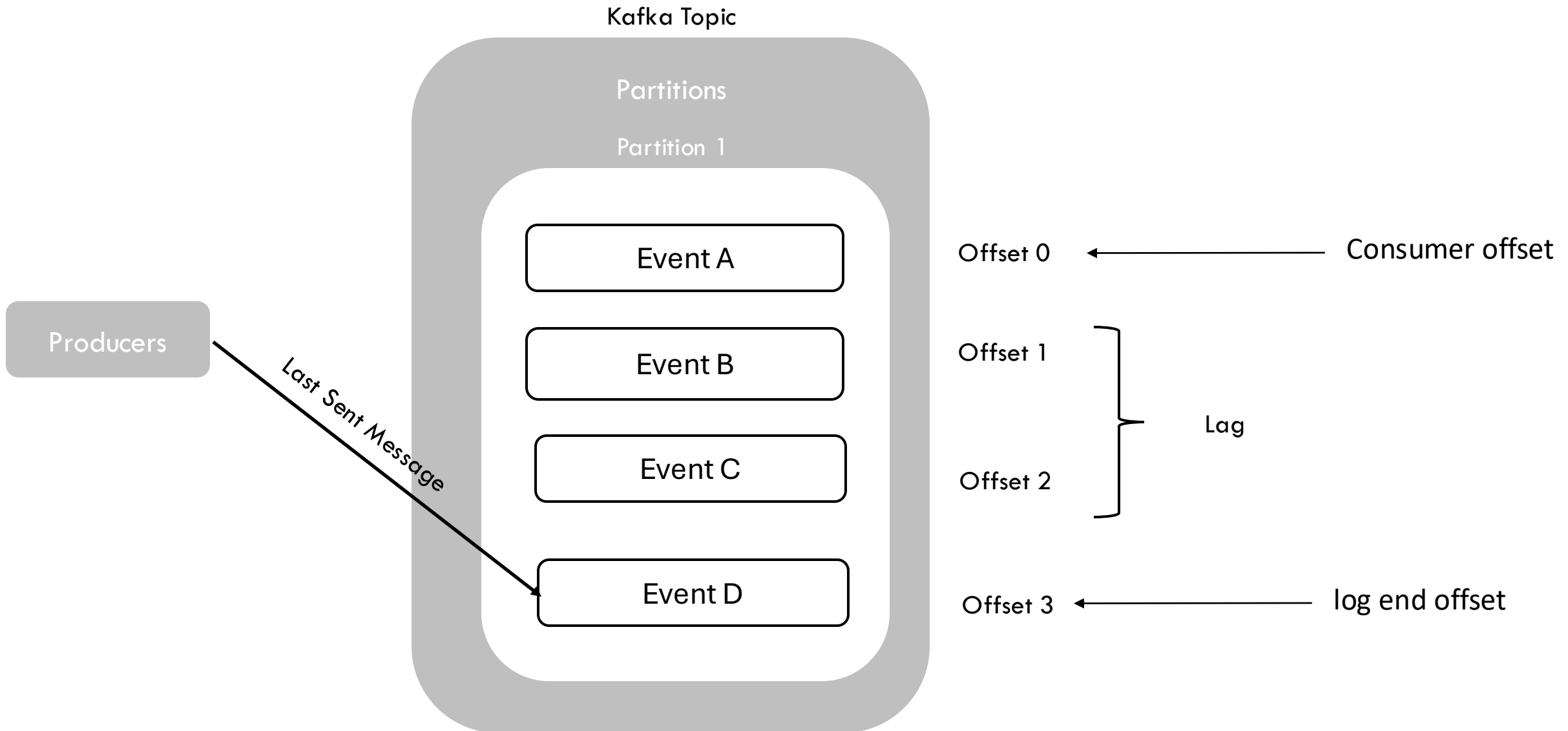
C1

C2

C3



Lag



Pattern-1



Pattern-2



